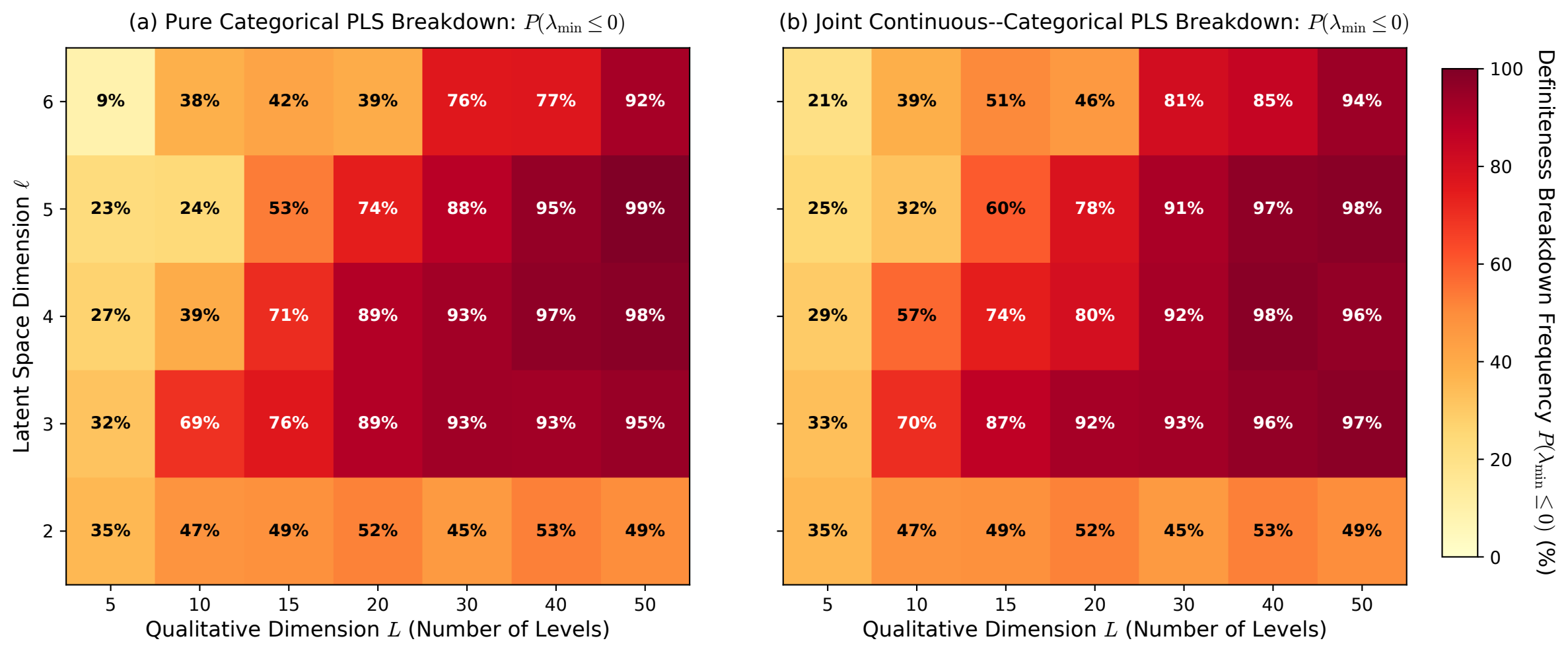

(a) Pure Categorical PLS Breakdown: $P(\lambda_{\min} \leq 0)$
(b) Joint Continuous--Categorical PLS Breakdown: $P(\lambda_{\min} \leq 0)$
Latent Space Dimension ℓ
6
5
4
3
2
9% 38% 42% 39% 76% 77% 92%
23% 24% 53% 74% 88% 95% 99%
27% 39% 71% 89% 93% 97% 98%
32% 69% 76% 89% 93% 93% 95%
35% 47% 49% 52% 45% 53% 49%
21% 39% 51% 46% 81% 85% 94%
25% 32% 60% 78% 91% 97% 98%
29% 57% 74% 80% 92% 98% 96%
33% 70% 87% 92% 93% 96% 97%
35% 47% 49% 52% 45% 53% 49%
5 10 15 20 30 40 50
Qualitative Dimension L (Number of Levels)
5 10 15 20 30 40 50
Qualitative Dimension L (Number of Levels)
100
80
60
40
20
0
Definiteness Breakdown Frequency $P(\lambda_{\min} \leq 0)$ (%)